In [ ]:
```
pip install keras
```

```
Requirement already satisfied: keras in /usr/local/lib/python3.13/dist-pa
ckages (3.13.2)
Requirement already satisfied: absl-py in /usr/local/lib/python3.13/dist-
packages (from keras) (1.4.0)
Requirement already satisfied: numpy in /usr/local/lib/python3.13/dist-pa
ckages (from keras) (2.1.3)
Requirement already satisfied: rich in /usr/local/lib/python3.13/dist-pac
kages (from keras) (13.9.4)
Requirement already satisfied: namex in /usr/local/lib/python3.13/dist-pa
ckages (from keras) (0.1.0)
Requirement already satisfied: h5py in /usr/local/lib/python3.13/dist-pac
kages (from keras) (3.14.0)
Requirement already satisfied: optree in /usr/local/lib/python3.13/dist-p
ackages (from keras) (0.20.0)
Requirement already satisfied: ml-dtypes in /usr/local/lib/python3.13/dis
t-packages (from keras) (0.6.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.13/dis
t-packages (from keras) (26.3)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/li
b/python3.13/dist-packages (from optree->keras) (4.16.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/py
thon3.13/dist-packages (from rich->keras) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/
python3.13/dist-packages (from rich->keras) (2.21.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.13/di
st-packages (from markdown-it-py>=2.2.0->rich->keras) (0.1.2)
```

In [ ]:
```
pip install tensorflow
```

```
Requirement already satisfied: tensorflow in /usr/local/lib/python3.13/di
st-packages (2.21.0)
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.1
3/dist-packages (from tensorflow) (1.4.0)
Requirement already satisfied: astunparse>=1.6.0 in /usr/local/lib/python
3.13/dist-packages (from tensorflow) (1.6.3)
Requirement already satisfied: flatbuffers>=25.9.23 in /usr/local/lib/pyt
hon3.13/dist-packages (from tensorflow) (25.12.19)
Requirement already satisfied: gast!=0.5.0,!=0.5.1,!=0.5.2,>=0.2.1 in /us
r/local/lib/python3.13/dist-packages (from tensorflow) (0.7.0)
Requirement already satisfied: google_pasta>=0.1.1 in /usr/local/lib/pyth
on3.13/dist-packages (from tensorflow) (0.2.0)
Requirement already satisfied: libclang>=13.0.0 in /usr/local/lib/python
3.13/dist-packages (from tensorflow) (18.1.1)
Requirement already satisfied: opt_einsum>=2.3.2 in /usr/local/lib/python
3.13/dist-packages (from tensorflow) (3.4.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.13/dis
t-packages (from tensorflow) (26.3)
Requirement already satisfied: protobuf<8.0.0,>=6.31.1 in /usr/local/lib/
python3.13/dist-packages (from tensorflow) (6.33.6)
Requirement already satisfied: requests<3,>=2.21.0 in /usr/local/lib/pyth
on3.13/dist-packages (from tensorflow) (2.32.4)
Requirement already satisfied: setuptools in /usr/local/lib/python3.13/di
st-packages (from tensorflow) (80.10.2)
Requirement already satisfied: six>=1.12.0 in /usr/local/lib/python3.13/d
ist-packages (from tensorflow) (1.17.0)
Requirement already satisfied: termcolor>=1.1.0 in /usr/local/lib/python
3.13/dist-packages (from tensorflow) (3.3.0)
Requirement already satisfied: typing_extensions>=3.6.6 in /usr/local/li
b/python3.13/dist-packages (from tensorflow) (4.16.0)
Requirement already satisfied: wrapt>=1.11.0 in /usr/local/lib/python3.1
3/dist-packages (from tensorflow) (2.5.0)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in /usr/local/lib/pyth
on3.13/dist-packages (from tensorflow) (1.84.0)
Requirement already satisfied: keras>=3.12.0 in /usr/local/lib/python3.1
3/dist-packages (from tensorflow) (3.13.2)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.1
3/dist-packages (from tensorflow) (2.1.3)
Requirement already satisfied: h5py<3.15.0,>=3.11.0 in /usr/local/lib/pyt
hon3.13/dist-packages (from tensorflow) (3.14.0)
Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in /usr/local/lib/
python3.13/dist-packages (from tensorflow) (0.6.0)
Requirement already satisfied: wheel<1.0,>=0.23.0 in /usr/local/lib/pytho
n3.13/dist-packages (from astunparse>=1.6.0->tensorflow) (0.48.0)
Requirement already satisfied: rich in /usr/local/lib/python3.13/dist-pac
kages (from keras>=3.12.0->tensorflow) (13.9.4)
Requirement already satisfied: namex in /usr/local/lib/python3.13/dist-pa
ckages (from keras>=3.12.0->tensorflow) (0.1.0)
Requirement already satisfied: optree in /usr/local/lib/python3.13/dist-p
ackages (from keras>=3.12.0->tensorflow) (0.20.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/li
b/python3.13/dist-packages (from requests<3,>=2.21.0->tensorflow) (3.4.9)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.13/
dist-packages (from requests<3,>=2.21.0->tensorflow) (3.20)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/pytho
```

```
n3.13/dist-packages (from requests<3,>=2.21.0->tensorflow) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/pytho
n3.13/dist-packages (from requests<3,>=2.21.0->tensorflow) (2026.7.22)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/py
thon3.13/dist-packages (from rich->keras>=3.12.0->tensorflow) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/
python3.13/dist-packages (from rich->keras>=3.12.0->tensorflow) (2.21.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.13/di
st-packages (from markdown-it-py>=2.2.0->rich->keras>=3.12.0->tensorflow)
(0.1.2)
```

In [ ]:
```
pip install contractions
```

```
Collecting contractions
  Downloading contractions-0.1.73-py2.py3-none-any.whl.metadata (1.2 kB)
Collecting textsearch>=0.0.21 (from contractions)
  Downloading textsearch-0.0.24-py2.py3-none-any.whl.metadata (1.2 kB)
Collecting anyascii (from textsearch>=0.0.21->contractions)
  Downloading anyascii-0.3.3-py3-none-any.whl.metadata (1.6 kB)
Collecting pyahocorasick (from textsearch>=0.0.21->contractions)
  Downloading pyahocorasick-2.3.1-cp313-cp313-manylinux2014_x86_64.manyli
nux_2_17_x86_64.whl.metadata (13 kB)
Downloading contractions-0.1.73-py2.py3-none-any.whl (8.7 kB)
Downloading textsearch-0.0.24-py2.py3-none-any.whl (7.6 kB)
Downloading anyascii-0.3.3-py3-none-any.whl (345 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 345.1/345.1 kB 8.4 MB/s eta
0:00:00
Downloading pyahocorasick-2.3.1-cp313-cp313-manylinux2014_x86_64.manylinu
x_2_17_x86_64.whl (114 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 114.8/114.8 kB 9.0 MB/s eta
0:00:00
Installing collected packages: pyahocorasick, anyascii, textsearch, contr
actions
Successfully installed anyascii-0.3.3 contractions-0.1.73 pyahocorasic
k-2.3.1 textsearch-0.0.24
```

In [ ]:
```
import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))
```

In [ ]:
```
from textblob import Word
import nltk
nltk.data.path.append("/kaggle/working/")
nltk.download('wordnet')
```

```
[nltk_data] Downloading package wordnet to /root/nltk_data...
```

Out[ ]:
```
True
```

In [ ]:
```
pip install --upgrade tensorflow
```

```
Requirement already satisfied: tensorflow in /usr/local/lib/python3.13/di
st-packages (2.21.0)
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.1
3/dist-packages (from tensorflow) (1.4.0)
Requirement already satisfied: astunparse>=1.6.0 in /usr/local/lib/python
3.13/dist-packages (from tensorflow) (1.6.3)
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hon3.13/dist-packages (from tensorflow) (25.12.19)
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Requirement already satisfied: opt_einsum>=2.3.2 in /usr/local/lib/python
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Requirement already satisfied: termcolor>=1.1.0 in /usr/local/lib/python
3.13/dist-packages (from tensorflow) (3.3.0)
Requirement already satisfied: typing_extensions>=3.6.6 in /usr/local/li
b/python3.13/dist-packages (from tensorflow) (4.16.0)
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on3.13/dist-packages (from tensorflow) (1.84.0)
Requirement already satisfied: keras>=3.12.0 in /usr/local/lib/python3.1
3/dist-packages (from tensorflow) (3.13.2)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.1
3/dist-packages (from tensorflow) (2.1.3)
Requirement already satisfied: h5py<3.15.0,>=3.11.0 in /usr/local/lib/pyt
hon3.13/dist-packages (from tensorflow) (3.14.0)
Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in /usr/local/lib/
python3.13/dist-packages (from tensorflow) (0.6.0)
Requirement already satisfied: wheel<1.0,>=0.23.0 in /usr/local/lib/pytho
n3.13/dist-packages (from astunparse>=1.6.0->tensorflow) (0.48.0)
Requirement already satisfied: rich in /usr/local/lib/python3.13/dist-pac
kages (from keras>=3.12.0->tensorflow) (13.9.4)
Requirement already satisfied: namex in /usr/local/lib/python3.13/dist-pa
ckages (from keras>=3.12.0->tensorflow) (0.1.0)
Requirement already satisfied: optree in /usr/local/lib/python3.13/dist-p
ackages (from keras>=3.12.0->tensorflow) (0.20.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/li
b/python3.13/dist-packages (from requests<3,>=2.21.0->tensorflow) (3.4.9)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.13/
dist-packages (from requests<3,>=2.21.0->tensorflow) (3.20)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/pytho
```

```
n3.13/dist-packages (from requests<3,>=2.21.0->tensorflow) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/pytho
n3.13/dist-packages (from requests<3,>=2.21.0->tensorflow) (2026.7.22)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/py
thon3.13/dist-packages (from rich->keras>=3.12.0->tensorflow) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/
python3.13/dist-packages (from rich->keras>=3.12.0->tensorflow) (2.21.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.13/di
st-packages (from markdown-it-py>=2.2.0->rich->keras>=3.12.0->tensorflow)
(0.1.2)
```

In [ ]:
```
pip install --upgrade gensim
```

```
Requirement already satisfied: gensim in /usr/local/lib/python3.13/dist-p
ackages (4.4.0)
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.1
3/dist-packages (from gensim) (2.1.3)
Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.13/
dist-packages (from gensim) (1.16.3)
Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python
3.13/dist-packages (from gensim) (8.0.2)
Requirement already satisfied: wrapt in /usr/local/lib/python3.13/dist-pa
ckages (from smart_open>=1.8.1->gensim) (2.5.0)
```

In [ ]:
```
import numpy as np
import pandas as pd
import contractions
```

In [ ]:
```
import gensim.downloader as api
```

In [ ]:
```
try:
    # Using on_bad_lines='skip' to skip rows with formatting issues like
    df = pd.read_csv('/content/IMDB Dataset.csv', on_bad_lines='skip')
except FileNotFoundError:
    print("Error: 'IMDB Dataset.csv' not found at /content/. Please uploa
```

In [ ]:
```
import seaborn as sns
import matplotlib.pyplot as plt
sns.set(style="white",font_scale=1.5)
sns.set(rc={"axes.facecolor":"#FFFAF0","figure.facecolor":"#FFFAF0"})
sns.set_context("poster",font_scale = .7)
```

In [ ]:
```
import contractions
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.preprocessing import LabelBinarizer
from nltk.corpus import stopwords
from nltk.stem.porter import PorterStemmer
from wordcloud import WordCloud,STOPWORDS
from nltk.stem import WordNetLemmatizer
from nltk.tokenize import word_tokenize,sent_tokenize
import spacy
import math
from gensim.models import Word2Vec
```

```
from sklearn.feature_extraction.text import TfidfVectorizer
from joblib import Parallel, delayed
import re,string,unicodedata
from sklearn.model_selection import train_test_split
from textblob import TextBlob
from scipy.sparse import lil_matrix
from tensorflow.keras.preprocessing.sequence import pad_sequences
from nltk import pos_tag

# Library for model building
from tensorflow.keras.regularizers import l2
import tensorflow
import keras
from keras import layers
from tensorflow.keras.preprocessing.text import Tokenizer
from keras.models import Sequential
#from attention import BahdanauAttention
from keras.layers import SimpleRNN,LSTM,GRU, Embedding, Dense, SpatialDro
from sklearn.metrics import accuracy_score
from tensorflow.keras.initializers import GlorotUniform
from tensorflow.keras.optimizers import Adam
#Library to overcome Warnings
import warnings
warnings.filterwarnings('ignore')
```

In [ ]:
```
import contractions
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.preprocessing import LabelBinarizer
from nltk.corpus import stopwords
from nltk.stem.porter import PorterStemmer
from wordcloud import WordCloud,STOPWORDS
from nltk.stem import WordNetLemmatizer
from nltk.tokenize import word_tokenize,sent_tokenize
import spacy
import math
from gensim.models import Word2Vec
from sklearn.feature_extraction.text import TfidfVectorizer
from joblib import Parallel, delayed
import re,string,unicodedata
from sklearn.model_selection import train_test_split
from textblob import TextBlob
from scipy.sparse import lil_matrix
from tensorflow.keras.preprocessing.sequence import pad_sequences
from nltk import pos_tag

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from tensorflow.keras.regularizers import l2
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from tensorflow.keras.preprocessing.text import Tokenizer
from keras.models import Sequential
#from attention import BahdanauAttention
from keras.layers import SimpleRNN,LSTM,GRU, Embedding, Dense, SpatialDro
```

```
from sklearn.metrics import accuracy_score
from tensorflow.keras.initializers import GlorotUniform
from tensorflow.keras.optimizers import Adam
#Library to overcome Warnings
import warnings
warnings.filterwarnings('ignore')
```

In [ ]:
```
df
```

Out[ ]:

	review	sentiment
0	One of the other reviewers has mentioned that ...	positive
1	A wonderful little production. The...	positive
2	I thought this was a wonderful way to spend ti...	positive
3	Basically there's a family where a little boy ...	negative
4	Petter Mattei's "Love in the Time of Money" is...	positive
...	...	...
49995	I thought this movie did a down right good job...	positive
49996	Bad plot, bad dialogue, bad acting, idiotic di...	negative
49997	I am a Catholic taught in parochial elementary...	negative
49998	I'm going to have to disagree with the previou...	negative
49999	No one expects the Star Trek movies to be high...	negative

50000 rows × 2 columns

In [ ]:
```
print('dataset shape: ',df.shape)
```

```
dataset shape:  (50000, 2)
```

In [ ]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 2 columns):
 #   Column     Non-Null Count  Dtype 
---  ------     --------------  ----- 
 0   review     50000 non-null  object
 1   sentiment  50000 non-null  object
dtypes: object(2)
memory usage: 781.4+ KB
```

In [ ]:
```
print("Duplicates in Dataset: ",df.duplicated().sum())
```

```
Duplicates in Dataset:  418
```

In [ ]:
```
df.drop_duplicates(inplace = True)
```

In [ ]:
```
missing_data = df.isnull().sum().to_frame().rename(columns={0:"Total No.
missing_data["% of Missing Values"] = round((missing_data["Total No. of M
missing_data
```

Out[ ]:

	Total No. of Missing Values	% of Missing Values
review	0	0.0
sentiment	0	0.0

In [ ]:
```
round(df.describe(include="O").T,2)
```

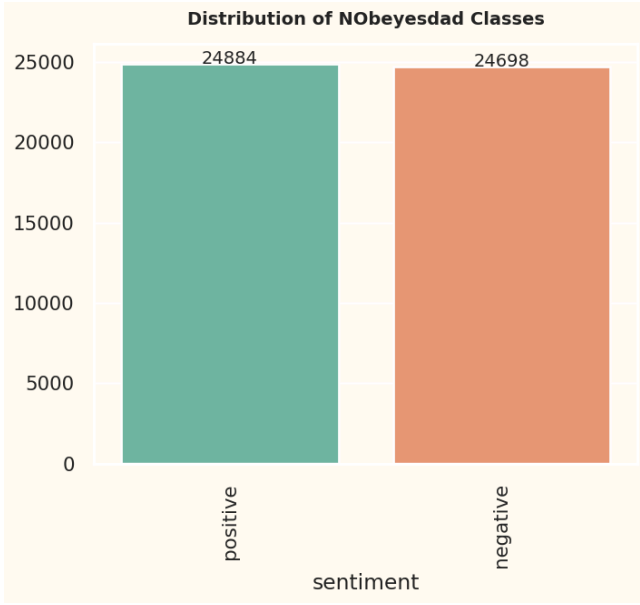
Out[ ]:

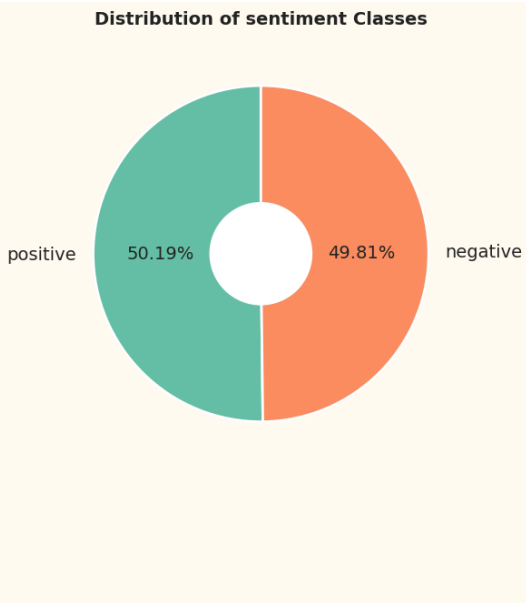
	count	unique	top	freq
review	49582	49582	No one expects the Star Trek movies to be high...	1
sentiment	49582	2	positive	24884

In [ ]:
```
classes = ['positive','negative']

plt.figure(figsize=(17, 6))
plt.subplot(1, 2, 1)
sentiment_counts = df["sentiment"].value_counts()
sns.barplot(x=sentiment_counts.index, y=sentiment_counts.values, palette=
plt.title("Distribution of NObeyesdad Classes", fontweight="black", size=
for i, v in enumerate(sentiment_counts.values):
    plt.text(i, v, v, ha="center", fontsize=14)
plt.xticks(rotation=90)

plt.subplot(1, 2, 2)
colors = sns.color_palette('Set2', len(classes))
plt.pie(sentiment_counts, labels=classes, autopct="%.2f%%", textprops={"s
        colors=colors, startangle=90)
center_circle = plt.Circle((0, 0), 0.3, fc='white')
fig = plt.gcf()
fig.gca().add_artist(center_circle)
plt.title("Distribution of sentiment Classes", fontweight="black", size=1
plt.show()
```


Distribution of NObeyesdad Classes
25000
20000
15000
10000
5000
0
24884
24698
positive
negative
sentiment


Distribution of sentiment Classes
positive
50.19%
49.81%
negative